

Algebraic Notation — Board-Focal ARA

This document specifies the mathematical structure used when the **Board is the focal (maximising) actor**. It is a Board-specific subset of the full `algebraic.md`; ASA-advice mode recursion is omitted but all utility functions are retained (they are consumed by the predictive distribution engine when modelling opponents).

1. Game tree structure

1.1 Node order

The tree is **not** a fixed linear sequence. Node traversal is conditional on game state, principally on whether the CEO is still present. The base sequence is:

$$D_0^{\text{ceo}} \rightarrow D_1 \rightarrow A_2 \rightarrow V \rightarrow M_{\text{agm}} \rightarrow D_4 \rightarrow D_{\text{rev}} \rightarrow R \rightarrow M_{\text{rev}} \rightarrow [\text{conditional } D_4 \rightarrow D_{\text{rev}}] \rightarrow \text{Terminal}$$

Key ordering principle: D_4 (CEO response) precedes D_{rev} (Board response). The CEO has initiative to resign before the Board decides whether to act.

Post-review conditional round: After R and M_{rev} , if `review_adverse = true` AND `CEO_present = true`, a second $D_4 \rightarrow D_{\text{rev}}$ round occurs. Otherwise, the game proceeds directly to Terminal.

Node	Type	Owner	Board role
D_0^{ceo}	Decision	CEO	Opponent (ARA)
D_1	Decision	Board	Focal (max)
A_2	Decision	ASA	Opponent (ARA)
V	Chance	Nature	MC integration
M_{agm}	Chance	Nature	Pass-through
D_4	Decision	CEO	Opponent (ARA)

D_{rev}	Decision	Board	Focal (max)
R	Chance	Nature	MC integration
M_{rev}	Chance	Nature	Pass-through
D_4'	Decision	CEO	Opponent (ARA)
D_{rev}'	Decision	Board	Focal (max)
Terminal	Terminal	—	Compute u_B

1.2 Action sets

Node	Actions	Feasibility
D_0^{ceo}	CEO_resign, CEO_stay	always
D_1	D0_minimal, D1_review, D3_ceo_transition	always (D3 requires CEO_present)
A_2	A2_no_strike, A2_rec_strike	always
D_4, D_4'	D4_stay, D4_resign, D4_negotiate_exit	CEO_present
$D_{\text{rev}}, D_{\text{rev}}'$	Drev_no_action, Drev_commission_review, Drev_sack_ceo	see below

Feasibility rules evaluated dynamically on the game state S :

Code	Condition
always	true
CEO_present	$S.\text{CEO_present} = \text{true}$
CEO_not_removed	$S.\text{CEO_removed} = \text{false}$
review_not_commissioned	$S.\text{review_commissioned} = \text{false}$
review_commissioned	$S.\text{review_commissioned} = \text{true}$

review_completed	$S.\text{review_completed} = \text{true}$
not_reviewed_yet	$S.\text{review_commissioned} = \text{false}$ $\wedge S.\text{review_completed} = \text{false}$
post_review_round	$S.\text{post_review_round} = \text{true}$
post_review_round_and_CEO_present	$S.\text{post_review_round}$ $\wedge S.\text{CEO_present}$
post_review_round_and_review_not_commissioned	$S.\text{post_review_round}$ $\wedge S.\text{review_commissioned} = \text{false}$

1.3 State transitions

The game state S is an immutable record:

S

$= (\text{CEO_present}, \text{review_commissioned}, \text{review_completed}, \text{CEO_removed}, \text{CEO_resigned_early}, r)$

Key transitions:

Action	Effect
CEO_resign	$\text{CEO_present} \leftarrow \text{false}, \text{CEO_removed} \leftarrow \text{true}, \text{CEO_resigned_early} \leftarrow \text{true}$
D3_ceo_transition, Drev_sack_ceo	$\text{CEO_present} \leftarrow \text{false}, \text{CEO_removed} \leftarrow \text{true}$
D1_review, Drev_commission_review	$\text{review_commissioned} \leftarrow \text{true}$
D4_resign, D4_negotiate_exit	$\text{CEO_present} \leftarrow \text{false}, \text{CEO_removed} \leftarrow \text{true}$
adverse (at R)	$\text{review_adverse} \leftarrow \text{true}, \text{review_completed} \leftarrow \text{true}, \text{post_review_round} \leftarrow \text{CEO_present}$

no_adverse (at R)	review_adverse $\leftarrow$ false, review_completed $\leftarrow$ true
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1.4 Scenario branching at D_0^{ceo}

The solver evaluates both pre-game scenarios separately:

$$\text{scenario} \in \{\text{ceo_stayed}, \text{ceo_resigned}\}$$

The probability of each scenario is computed via Bayesian-updated ARA predictive distribution over D_0^{ceo} (see §9.2):

$$\Pr_B(\text{CEO_resign}) = \frac{\alpha_0 + n_{\text{resign}}}{\alpha_0 + \beta_0 + N}$$

where $\alpha_0 = 12.5$, $\beta_0 = 0.5$ (Beta prior from 12 Australian no-remorse CEO observations), and n_{resign} is ARA soft evidence.

1.5 Conditional branching structure

When $D_0^{\text{ceo}} = \text{CEO_stay}$ and $D_1 \neq \text{D3_ceo_transition}$, the full tree unfolds:

$$\begin{aligned} D_1 \rightarrow A_2 \rightarrow V \rightarrow M_{\text{agm}} \rightarrow D_4 \rightarrow D_{\text{rev}} \rightarrow R \rightarrow M_{\text{rev}} \\ \rightarrow \begin{cases} D_4' \rightarrow D_{\text{rev}}' \rightarrow \text{Terminal} & \text{if adverse} \wedge \text{CEO_present} \\ \text{Terminal} & \text{otherwise} \end{cases} \end{aligned}$$

When CEO is removed early ($D_0^{\text{ceo}} = \text{CEO_resign}$ or $D_1 = \text{D3_ceo_transition}$):

$$D_1 \rightarrow A_2 \rightarrow V \rightarrow M_{\text{agm}} \rightarrow [\text{D4 skip}] \rightarrow D_{\text{rev}} \rightarrow R \rightarrow M_{\text{rev}} \rightarrow \text{Terminal}$$

2. Belief checkpoint draws

Each checkpoint $c \in \{C_{\text{pre}}, C_0, C_1, C_2, C_3\}$ contains N draws (typically 500) from upstream Stan models:

$$\theta_c^{(i)} = (B_{\text{mkt}}^{(i)}, B_{\text{mgmt}}^{(i)}, \alpha_V^{(i)}, \gamma_A^{(i)}, \gamma_{AH}^{(i)}, \gamma_D^{(i)}, \sigma_V^{(i)}, r_1^{(i)}, r_2^{(i)}), \quad i = 1, \dots, N$$

Opponent parameter priors $\theta_j \sim p_B(\theta_j)$ for $j \in \{\text{ASA}, \text{CEO}\}$ are specified in `opponent_priors.xlsx`. Supported families: Normal, LogNormal, Beta, Uniform, Gamma.

3. Chance models

3.1 Vote model — logit-normal with headline interaction and crisis floor

$$B_{\text{agm}} = B_{\text{mkt}}^{(i)} + \gamma_A^{(i)} \cdot \mathbb{1}[A_2 = \text{rec_strike}] + \gamma_{AH}^{(i)} \cdot \mathbb{1}[A_2 = \text{rec_strike}] \cdot \mathbb{1}[\text{headline}] \\ + \gamma_D^{(i)} \cdot f(D_1)$$

Governance effect:

$$f(D_1) = \begin{cases} 0 & D_1 = \text{D0_minimal} \\ U(\ell_1, u_1) & D_1 = \text{D1_review} \\ U(\ell_3, u_3) & D_1 = \text{D3_ceo_transition} \end{cases}$$

Unbiased bounds: $(\ell_1, u_1) = (0, 1)$, $(\ell_3, u_3) = (-1, 0.5)$.

Vote fraction from logit-normal:

$$\text{logit}(V) \sim \mathcal{N}(\alpha_V^{(i)} + B_{\text{agm}}, \sigma_V^{(i)}), \quad V_{\text{logit}} = \text{expit}(\text{logit}(V))$$

Structural crisis floor (headline incidents only):

$$V_{\text{floor}} \sim \text{Beta}(50, 150), \quad V_{\text{final}} = \max(V_{\text{logit}}, V_{\text{floor}})$$

V_{floor} drawn once per belief draw (epistemic).

Derived indicators:

$$\text{strike} = \mathbb{1}[V_{\text{final}} \geq 0.25], \quad \text{overwhelming} = \mathbb{1}[V_{\text{final}} \geq 0.50]$$

Epistemic/aleatoric separation: Three epistemic quantities drawn once per belief draw:

1. $f(D_1) \sim U(\ell, u)$ — governance effect
2. $V_{\text{floor}} \sim \text{Beta}(50, 150)$ — crisis floor
3. κ_σ — sigma scale (when Board overconfidence bias is active)

3.2 Review findings — two-component model

Component 1: Outcome rating.

$$p_{\text{adverse}} \sim \text{Beta}(10,5), \quad \text{review_adverse} \sim \text{Bernoulli}(p_{\text{adverse}})$$

p_{adverse} drawn once per belief draw (epistemic). Based on Dirichlet(5,5,5) grouping negative + neutral as "adverse". Mean = 2/3.

Component 2: CAR.

$$\mu_f \sim t_4(-0.05, 0.03), \quad \sigma_f \sim |\mathcal{N}(0, 0.10)|, \quad \text{CAR} \sim t_3(\mu_f, \sigma_f)$$

The outcome rating and CAR are independent: adverse reviews can produce mildly positive CAR (market had priced in bad news) and vice versa.

If review not commissioned: CAR = 0, adverse = false.

3.3 Review direct cost

$$C_{\text{direct}} \sim \text{Gamma}(4.55, 4741), \quad \text{mean} \approx 9.6 \text{ bps}$$

Drawn once per scenario (epistemic).

4. Board overconfidence bias

When Board is focal, three cognitive biases are applied:

4.1 Overestimation on governance effects

$$f(D_1 = D1_{\text{review}}) \sim U(0.63, 1.0), \quad f(D_1 = D3) \sim U(-0.62, 0.5)$$

($\beta = 0.625$ midpoint overestimation.)

4.2 Overprecision on vote uncertainty

$$\phi_V = \sigma_V \cdot \kappa_\sigma, \quad \kappa_\sigma = 1/\sqrt{\kappa}, \quad \kappa = 3.5$$

Production default: $\kappa_\sigma = 0.53$.

4.3 Overestimation on review

CAR location:

$$\hat{\mu}_f = \mu_f + 0.03$$

Outcome rating: Board believes positive outcomes are more likely:

$$p_{\text{adverse}}^{\text{biased}} \sim \text{Beta}(10, 6.5), \quad \mathbb{E} = 0.606 \quad (\text{vs } 0.667 \text{ unbiased})$$

$$\hat{\beta}_{\text{positive}} = 5 \times (1 + 10 \times 0.03) = 6.5.$$

4.4 Bias propagation

All biases apply consistently in both the Board's own EU calculation and in rollout simulations for predictive distributions, producing self-consistent decision-making.

5. Utility functions

5.1 Terminal outcome

Z

$$= (d_1, a_2, d_{\text{rev}}, d_4, d_4', d_{\text{rev}}', V, \text{strike}, \text{overwhelming}, \text{CAR}, C_{\text{direct}}, \text{adverse}, \text{CEO_removed}, \text{CEO_resigned_early})$$

5.2 Board utility (focal)

Board minimises opposition and disruption. Define $\text{ceo_at_end} = \neg \text{CEO_removed} \wedge \neg \text{CEO_resigned_early}$.

$$\begin{aligned} u_B(Z) = & -w_{\text{early}} \cdot \mathbb{1}[\text{CEO_resigned_early}] \\ & -w_{\text{vote}} \cdot (V - 0.25)_+^2 \\ & -w_{\text{over}} \cdot \mathbb{1}[\text{overwhelming}] \\ & -w_{\text{spill}} \cdot V \cdot \mathbb{1}[\text{strike}] \\ & +w_{\text{CAR}} \cdot \text{CAR} \cdot \mathbb{1}[\text{review_commissioned}] \\ & -w_{\text{cost}} \cdot C_{\text{direct}} \cdot \mathbb{1}[\text{review_commissioned}] \\ & -w_{\text{impl}} \cdot (\mathbb{1}[d_1 = D3] + \mathbb{1}[d_{\text{rev}} = \text{sack}] + \mathbb{1}[d_{\text{rev}}' = \text{sack}]) \\ & -w_{\text{loss}} \cdot \mathbb{1}[\text{CEO_removed} \wedge \neg \text{CEO_resigned_early}] \\ & -w_{\text{rep}} \cdot \mathbb{1}[\text{overwhelming}] \\ & -w_{\text{spill2}} \cdot \mathbb{1}[\text{strike} \wedge \text{ceo_at_end}] \\ & -w_{\text{reg}} \cdot \mathbb{1}[\text{strike} \wedge \text{ceo_at_end}] \\ & -w_{\text{d1_liab}} \cdot \mathbb{1}[\text{overwhelming} \wedge d_1 = D0_minimal] \\ & -w_{\text{legal_d1}} \cdot \mathbb{1}[\text{strike} \wedge d_1 = D0_minimal] \\ & -w_{\text{legal_rev}} \cdot \mathbb{1}[\text{strike} \wedge \text{ceo_at_end}] \\ & -w_{\text{adverse_ceo}} \cdot \mathbb{1}[\text{review_commissioned} \wedge \text{adverse} \wedge \text{ceo_at_end}] \end{aligned}$$

Legal and regulatory terms (last six lines):

Term	Trigger	Legal basis	Default
w_{spill2}	strike $\wedge$ ceo_at_end	Corporations Act 2001 s.250V: second-strike Board spill	8.0
w_{reg}	strike $\wedge$ ceo_at_end	ASIC director banning, personal fines	5.0
$w_{\text{d1_liab}}$	overwhelming $\wedge$ D0_minimal	Board inaction despite > 50% vote	4.0
$w_{\text{legal_d1}}$	strike $\wedge$ D0_minimal	Class actions, ACCC/ASIC penalties (D1 inaction)	3.0
$w_{\text{legal_rev}}$	strike $\wedge$ ceo_at_end	Company legal exposure (CEO retained post-strike)	2.0
$w_{\text{adverse_ceo}}$	review_commissioned $\wedge$ adverse $\wedge$ ceo_at_end	Liability from retaining CEO after adverse review	5.0

Full default weights:

Parameter	Symbol	Default
early_ceo_departure_cost	w_{early}	0.5
vote_penalty_weight	w_{vote}	2.0
overwhelming_penalty_weight	w_{over}	3.0
spill_risk_weight	w_{spill}	2.5
review_car_weight	w_{CAR}	15.0
review_direct_cost_weight	w_{cost}	15.0
implementation_cost_sack	w_{impl}	1.0
ceo_loss_cost	w_{loss}	1.5
reputational_spill_weight	w_{rep}	1.0

second_strike_spill_penalty	w_{spill2}	8.0
board_regulatory_liability	w_{reg}	5.0
board_d1_liability	$w_{\text{d1_liab}}$	4.0
qantas_legal_d1_penalty	$w_{\text{legal_d1}}$	3.0
qantas_legal_d_rev_penalty	$w_{\text{legal_rev}}$	2.0
adverse_review_ceo_present_penalty	$w_{\text{adverse_ceo}}$	5.0

5.3 ASA utility (opponent — used in predictive distribution)

$$\begin{aligned}
u_A(Z) = & w_{\text{early}}^A \cdot \mathbb{1}[\text{CEO_resigned_early}] \\
& + w_{\text{vote}}^A \cdot V \\
& + w_{\text{over}}^A \cdot \mathbb{1}[\text{overwhelming}] \\
& + w_{\text{removal}}^A \cdot \mathbb{1}[\text{CEO_removed} \wedge \neg \text{CEO_resigned_early}] \\
& - w_{\text{CAR}}^A \cdot \text{CAR} \cdot \mathbb{1}[\text{review_commissioned}] \\
& - w_{\text{mob}}^A \cdot \mathbb{1}[a_2 = \text{rec_strike}] \\
& + w_{\text{rep}}^A \cdot (V - 0.25)_+ \\
& + w_{\text{align}}^A \cdot \mathbb{1}[a_2 = \text{rec_strike} \wedge \text{strike}]
\end{aligned}$$

Default weights:

Parameter	Symbol	Default
early_ceo_departure_reward	w_{early}^A	2.0
vote_reward_weight	w_{vote}^A	2.0
overwhelming_reward_weight	w_{over}^A	2.0
ceo_removal_reward	w_{removal}^A	3.0
review_car_weight	w_{CAR}^A	15.0
mobilisation_cost	w_{mob}^A	0.3
reputational_gain_weight	w_{rep}^A	1.0

market_alignment_bonus	w_{align}^A	1.5
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5.4 CEO utility (opponent — used in predictive distribution)

$$u_C(Z) = U_{\text{money}}(W) - \lambda_D \cdot D_{\text{raw}}$$

CRRA with Kahneman–Tversky loss aversion:

$$\text{CRRA}(W) = \begin{cases} W^{1-\gamma}/(1-\gamma) & \gamma \neq 1 \\ \ln(W) & \gamma = 1 \end{cases}$$

$$U_{\text{money}}(W) = \begin{cases} \text{CRRA}(W) & W \geq W_{\text{ref}} \\ \lambda \cdot \text{CRRA}(W) - (\lambda - 1) \cdot \text{CRRA}(W_{\text{ref}}) & W < W_{\text{ref}} \end{cases}$$

Resign path: $W = W_{\text{resign}}, D_{\text{raw}} = D_{\text{resign}}$.

Stay path — wealth:

$$W = \begin{cases} W_{\text{stay_kept}} & \text{CEO not removed} \\ \frac{1}{2}(W_{\text{stay_sacked}} + W_{\text{stay_kept}}) & \mathcal{d}_4 = \text{negotiate_exit} \\ 1.3 \cdot W_{\text{stay_sacked}} & \mathcal{d}_4 = \text{resign} \\ W_{\text{stay_sacked}} & \text{otherwise forced removal} \end{cases}$$

Stay path — disutility (departure-mode-dependent):

$$D_{\text{raw}} = D_{\text{stay}} + D_{\text{departure}}(\mathcal{d}_4) \cdot \mathbb{1}[\text{CEO_removed}] + D_{\text{agm}} \cdot \mathbb{1}[V > 0.25] + D_{\text{disgrace}} \cdot \mathbb{1}[\text{overwhelming}] + D_{\text{adverse_review}} \cdot \mathbb{1}[\text{review_commissioned} \wedge \text{adverse}]$$

$$D_{\text{departure}}(\mathcal{d}_4) = \begin{cases} D_{\text{negotiate}} = 45 & \mathcal{d}_4 = \text{negotiate_exit} \\ D_{\text{resign_late}} = 60 & \mathcal{d}_4 = \text{resign} \\ D_{\text{sacked}} = 100 & \text{otherwise Board fires CEO} \end{cases}$$

$\mathcal{d}_4$ is the effective CEO departure action (d_4' overrides d_4 if the CEO acted post-review).

Default parameters:

Parameter	Default
γ	1.5

$\lambda = \lambda_D$	2.25
W_{ref}	16.0
W_{resign}	8.0
$W_{\text{stay_kept}}$	7.0
$W_{\text{stay_sacked}}$	1.5
D_{resign}	40.0
D_{stay}	25.0
D_{sacked}	100.0
$D_{\text{resign_late}}$	60.0
$D_{\text{negotiate}}$	45.0
D_{agm}	30.0
D_{disgrace}	30.0
$D_{\text{adverse_review}}$	10.0

6. ARA predictive distributions (Board perspective)

6.1 General form

At an opponent decision node X owned by player $j \in \{\text{ASA}, \text{CEO}\}$:

$$p_B(X = x \mid h) = \frac{1}{K} \sum_{k=1}^K \mathbb{1} \left[x \in \arg \max_{x' \in \mathcal{X}(h)} \Psi_j(x'; h, \theta_j^{(k)}) \right]$$

- $K = 200$ opponent parameter samples: $\theta_j^{(k)} \sim p_B(\theta_j)$

- $\Psi_j(x; h, \theta_j) = \frac{1}{R} \sum_{r=1}^R u_j(Z^{(r)}; \theta_j), R = 20$

Rollouts use fixed policies for non-evaluated actors and the Board's (possibly biased) model of Nature.

6.2 Fixed policies (Level-1 rollout fallbacks)

Board at D_1 : $\pi_B^{\text{fix}}(D_1) = \text{D0_minimal}$

Board at $D_{\text{rev}} / D_{\text{rev}}'$:

$$\pi_B^{\text{fix}}(D_{\text{rev}} | h) = \begin{cases} \text{Drev_sack_ceo} & V_{\%} \geq \tau_{\text{sack}} \wedge \text{CEO_present} \\ \text{Drev_commission_review} & V_{\%} \geq \tau_{\text{review}} \wedge \text{review_not_commissioned} \\ \text{Drev_no_action} & \text{otherwise} \end{cases}$$

$\tau_{\text{sack}} = 0.25, \tau_{\text{review}} = 0.25$ (from policy_parameters sheet).

ASA at A_2 — Beta-Binomial:

Board action	Posterior	Mean
D0_minimal	Beta(46,4)	0.920
D1_review / CEO resigned	Beta(44,4)	0.917
D3_ceo_transition	Beta(43,4)	0.914

$$p_{\text{strike}} \sim \text{Beta}(\alpha_d, \beta_d), \quad A_2 \sim \text{Bernoulli}(p_{\text{strike}})$$

CEO at D_0^{ceo} : $\pi_C^{\text{fix}} = \text{CEO_stay}$

CEO at D_4 / D_4' :

$$\pi_C^{\text{fix}}(D_4) = \begin{cases} \text{D4_resign} & V_{\%} \geq 0.40 \wedge \text{feasible} \\ \text{D4_stay} & \text{otherwise} \end{cases}$$

6.3 Level-2 recursion

At Level-2, opponents model the Board strategically:

- ASA models Board: $p_{\text{ASA}}^{(1)}(D_1 = d | h)$ replaces π_B^{fix} at D_1
- CEO models Board: $p_{\text{CEO}}^{(1)}(D_{\text{rev}} = d | h)$ replaces π_B^{fix} at D_{rev}

Nested predictions use reduced K_{nested} and R_{nested} .

7. Board-focal tree recursion

7.1 Node dispatch

Node type	Owner	Operation
Decision	Board	max over feasible actions
Decision	ASA / CEO	Σ weighted by $p_B(\cdot)$ or fixed policy
Chance (V)	Nature	MC average over M_V vote samples
Chance (R)	Nature	MC average over M_R review samples
Chance ($M_{\text{agm}}, M_{\text{rev}}$)	Nature	Pass-through
Terminal	—	Compute $u_B(Z)$

7.2 Full recursion (ceo_stayed scenario)

$$U_B^{(D_1)}(h_0) = \max_{d_1 \in \mathcal{D}_1} U_B^{(A_2)}(h_0 \cup \{D_1 = d_1\})$$

$$U_B^{(A_2)}(h) = \sum_{a \in \mathcal{A}_2} p_B(A_2 = a \mid h) \cdot U_B^{(V)}(h \cup \{A_2 = a\})$$

$$U_B^{(V)}(h) = \frac{1}{M_V} \sum_m U_B^{(D_4)}(h \cup \{V_m\})$$

M_{agm} is a pass-through.

CEO responds (opponent — predictive or skip if absent):

$$U_B^{(D_4)}(h) = \begin{cases} \sum_{d_4} p_B(D_4 = d_4 \mid h) \cdot U_B^{(D_{\text{rev}})}(h \cup \{D_4 = d_4\}) & \text{if CEO_present} \\ U_B^{(D_{\text{rev}})}(h) & \text{if CEO absent} \end{cases}$$

Board responds (focal — maximise):

$$U_B^{(D_{\text{rev}})}(h) = \max_{d \in \mathcal{D}_{\text{rev}}(S)} U_B^{(R)}(h \cup \{D_{\text{rev}} = d\})$$

Review findings:

$$U_B^{(R)}(h) = \frac{1}{M_R} \sum_m U_B^{(\text{post-review})}(h \cup \{R_m, C_{\text{direct}}\})$$

M_{rev} is a pass-through.

Post-review conditional round:

$$U_B^{(\text{post-review})}(h) = \begin{cases} U_B^{(D_4')} (h) & \text{if review_adverse} \wedge \text{CEO_present} \\ u_B(Z(h, S)) & \text{otherwise} \end{cases}$$

$$U_B^{(D_4')} (h) = \sum_{d_4} p_B(D_4 = d_4 \mid h) \cdot U_B^{(D_{\text{rev}})}(h \cup \{D_4 = d_4\})$$

$$U_B^{(D_{\text{rev}})}(h) = \max_{d \in \mathcal{D}_{\text{rev}}(S)} u_B(Z(h \cup \{D_{\text{rev}} = d\}, S))$$

7.3 Board optimal action

$$d_1^* \in \arg \max_{d_1 \in \mathcal{D}_1} \frac{1}{N} \sum_{i=1}^N U_B^{(A_2)}(h_0 \cup \{D_1 = d_1\}; \theta^{(i)})$$

7.4 Predictive distributions required

Distribution	Node	Opponent	Purpose
$p_B(A_2 \mid h)$	A_2	ASA	Strike recommendation
$p_B(D_4 \mid h)$	D_4	CEO	Post-AGM response
$p_B(D_4' \mid h)$	D_4'	CEO	Post-review response (same model)

8. Mode configurations (Board only)

Mode	ASA model	CEO model	Level

Board Mode	ARA	ARA	1
Board L2	ARA	ARA	2
Board (ASA=Policy)	Policy	ARA	1

Level-2 strategic counterparts: ASA models Board, CEO models Board.

9. Solver orchestration

9.1 Pipeline

For checkpoint c , scenario s :

1. Load BeliefBundle(c), utility weights, policy parameters, overconfidence bias.
2. Construct engine: ChanceModels, PredictiveDistribution($K = 200, R = 20$), TreeEvaluator($M_V = 50, M_R = 20$).
3. Set initial state: $S_0 = \text{for_scenario}(s)$.
4. For each $d_1 \in \mathcal{D}_1(S_0)$: $\text{EU}(d_1) = \frac{1}{N} \sum_{i=1}^N U_B^{(A_2)}(\{D_1 = d_1\}; \theta^{(i)})$.
5. $d_1^* = \underset{d_1}{\text{argmax}} \text{EU}(d_1)$.

9.2 DO_ceo Bayesian prediction

Prior: $p_{\text{resign}} \sim \text{Beta}(12.5, 0.5)$, $\mathbb{E} = 0.962$ (Jeffreys + 12 Australian CEO resignations).

ARA evidence: Level-2 predictive over N belief draws yields soft pseudo-counts $n_{\text{resign}}, n_{\text{stay}}$.

Posterior:

$$\Pr_B(\text{CEO_resign}) = \frac{12.5 + n_{\text{resign}}}{13.0 + N}$$

Computed with reduced budget: $K_{d0} = 50, R_{d0} = 10$.

9.3 Scenario solver

1. $\Pr_B(D_0^{\text{ceo}}) = \text{predict_d0_ceo}()$
2. For each $s \in \{\text{ceo_stayed}, \text{ceo_resigned}\}$: solve and attach $\Pr_B$.

9.4 Parallelisation

Each (d_1, i) pair submitted to ProcessPoolExecutor. Workers initialised once per checkpoint.

10. Computational budget

Parameter	Symbol	Default
Belief draws	N	500 (or --n_draws)
Opponent parameter samples	K	200
Stochastic rollouts	R	20
Vote MC samples	M_V	50
Review MC samples	M_R	20
D0_ceo opponent samples	K_{d0}	50
D0_ceo rollouts	R_{d0}	10

Total tree evaluations per solve: $|\mathcal{D}_1| \times N$. Total rollouts per predictive call: $K \times |\mathcal{A}| \times R$.